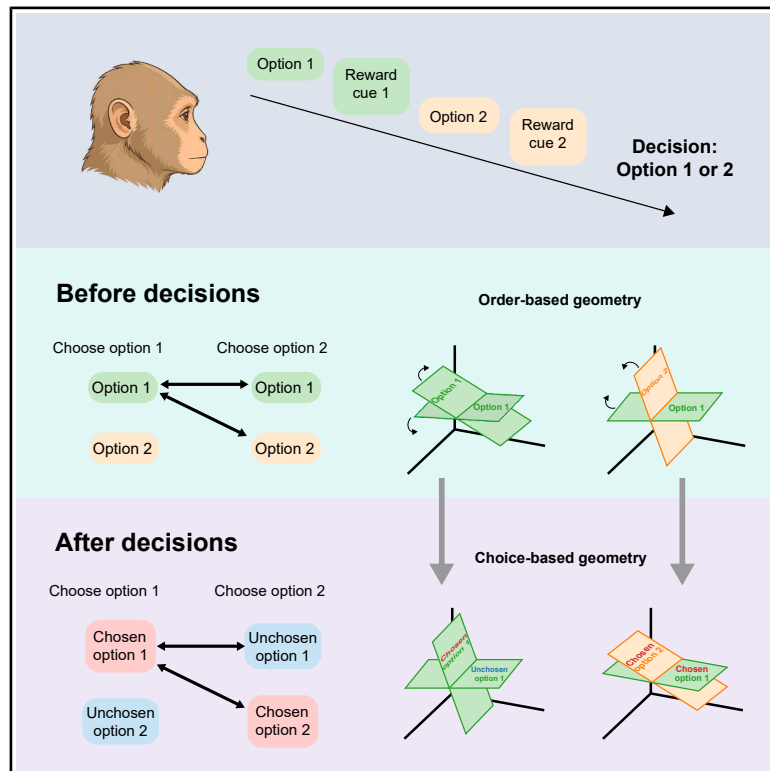


Neural subspace reorganization reflects value-based decision-making

Graphical abstract



Authors

Huidi Li, Nikolaos Chrysanthidis, Scott L. Brincat, Jonas Rose, Earl K. Miller

Correspondence

ekmiller@mit.edu

In brief

Biological sciences; Neuroscience; Behavioral neuroscience; Cognitive neuroscience

Highlights

- Prefrontal population geometry dynamically reorganized as decisions unfolded
- Before decisions, options were held in separate subspaces based on presented order
- After decisions, options were reorganized into separate subspaces based on choice
- Chosen option subspaces were aligned, facilitating efficient action selection



Article

Neural subspace reorganization reflects value-based decision-making

Huidi Li,¹ Nikolaos Chrysanthidis,¹ Scott L. Brincat,¹ Jonas Rose,^{1,2} and Earl K. Miller^{1,3,*}¹The Picower Institute for Learning and Memory, Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology, Cambridge, MA 02139, USA²Neural Basis of Learning, Institute of Cognitive Neuroscience, Faculty of Psychology, Ruhr University Bochum, Bochum, Germany³Lead contact*Correspondence: ekmiller@mit.edu<https://doi.org/10.1016/j.isci.2026.117492>

SUMMARY

Value-based decision-making requires maintaining and comparing multiple options to guide behavior. While the neural circuits supporting these decisions are increasingly understood, how decisions reshape option representations remains unclear. We recorded from lateral prefrontal cortex (LPFC) while non-human primates held two sequentially presented option-value pairs in working memory and selected the higher value option. We found that, once a decision could be made, representations of the chosen and unchosen options rotated into orthogonal neural subspaces, while the chosen representation expanded. Before the decision, representations were organized by presentation order. After the decision, the chosen option, regardless of presentation order, was transformed into a common subspace aligned with the prescribed action. These findings suggest that value-based decisions reorganize LPFC population activity to selectively represent the chosen option, enabling efficient downstream readout for action selection.

INTRODUCTION

To make an economic choice, we must compare the values of goods available and choose the goods with greater value. The process of making this choice can be broken down into multiple steps including representing available options, assigning values to them, comparing those values, and selecting the option with greater value.^{1–5} Previous research has studied how values are encoded and compared to make a choice.^{6–9} But it remains less well understood how multiple options are simultaneously represented in working memory and how these representations are transformed and reorganized by value-based choices.

Lateral prefrontal cortex (LPFC) is a likely candidate site for linking options, values, and actions, and for selecting actions based on their associated values.^{10–13} Multiple potential options often need to be maintained in working memory during value assignment. The LPFC is also implicated in encoding both choice values and action plans. It receives direct input from the orbitofrontal cortex,^{14,15} which is involved in processing reward value and other economic factors.^{1,2,16} It also has dense projections to motor systems.^{17–19} Neural activity in the LPFC has been shown to reflect the transformation of value into action.¹¹

Many recent studies have shown that population spiking activity in LPFC,^{20–29} as well as other frontal areas,^{30–33} exhibits structured correlations across neurons. As a result, neural activity tends to be restricted to low-dimensional “subspaces” within the high-dimensional space of all possible population activity patterns.^{34–36} Structuring activity within such subspaces is thought to

flexibly organize information processing^{20–25,31,37–40} and communication.^{41,42}

Motivated by these observations, we recorded from LPFC while non-human primates (NHPs) performed a value-based selection task. To segregate option representations before and after decisions, we used a novel task paradigm that temporally dissociated presentation of options from their value assignment. We examined the representational geometry of the options before and after their relative values were assigned and a decision could be made. We found that, before the decision, the two potential options were held in near-orthogonal subspaces. Once the decision was made, they were transformed to different subspaces that were now orthogonalized by whether their associated actions were chosen vs. unchosen, and the representation of the chosen action was expanded. Chosen actions were encoded within the same subspace, regardless of the initial presentation order of their associated option, facilitating their readout to generate behavior. These results show that organized changes in representational geometry of options support value-based decision-making.

RESULTS

Behavior and analysis

In each trial (Figure 1A), NHPs were presented with a series of two spatial targets (T1 and T2). Each was followed by a reward cue (R1 and R2, respectively) whose color signaled the reward value available if the preceding target was chosen. After the two targets and their values were presented, there was a



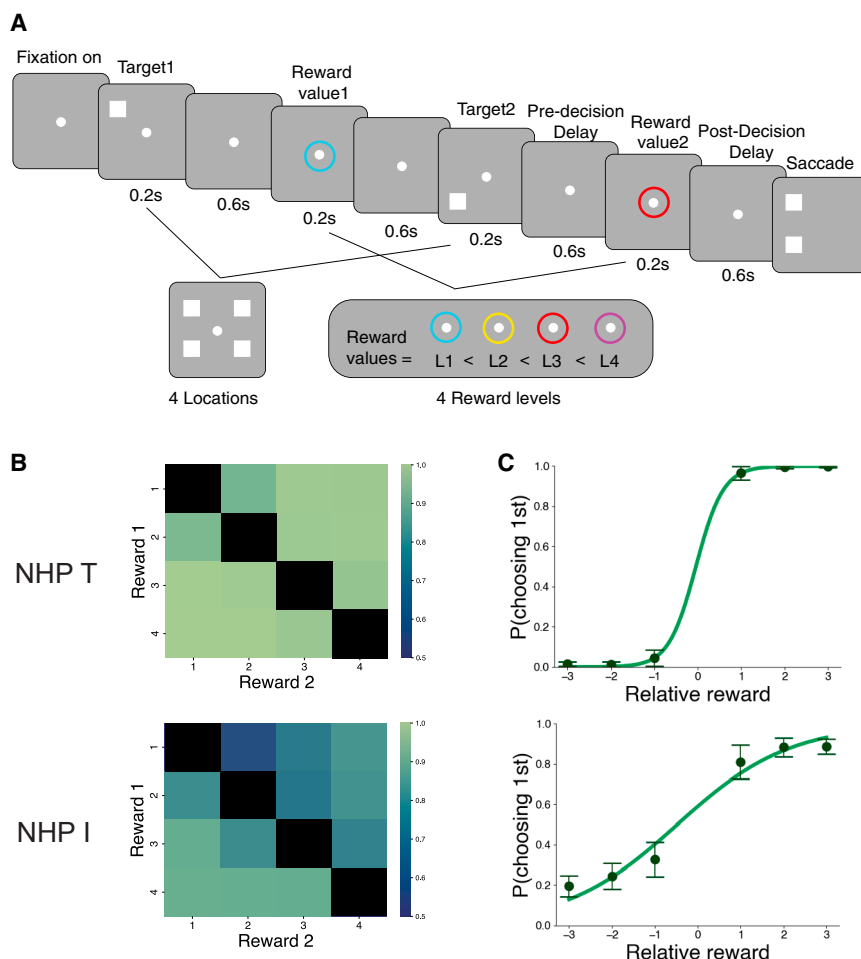


Figure 1. Experimental paradigm and behavioral results

(A) Time course of a task trial, proceeding from left to right. Two spatial targets (T1 and T2) were shown, paired with abstract cues (R1 and R2) signaling the reward value available if the preceding target was chosen. The NHPs' task was to choose the target associated with the higher reward.

(B) Probability of choosing the high-reward target for NHP T (top) and NHP I (bottom) across all combinations of first and second reward values. Values along the diagonal—where the same reward was offered for both targets—were not tested.

(C) Probability (mean $\pm$ SD across sessions) of choosing the first presented target at different relative reward levels (first-target reward–second-target reward). Both NHPs reliably chose the higher-value target across all pairs of first and second reward values.

value was known (R2), NHPs needed to hold both targets in memory (pre-decision delay). Then after R2, they could compare the two targets' reward values and make a decision (post-decision delay). Our data analysis was based on the spiking activity of neurons in LPFC for two NHPs (NHP T: 316 neurons; NHP I: 422 neurons).

LPFC encoded both reward and target location information

We examined how reward and target information evolved during the decision process. We trained a linear discriminant

“post-decision delay”. The NHPs had all the information needed to decide which target to choose. But they had to wait for a “go” cue (the reappearance of the two targets) before choosing one by saccading to it. Both NHPs most often chose the higher-value target over the lower value target (97.77% for NHP T; 78.56% for NHP I). Both NHPs reliably chose the higher value target across all pairs of first and second reward values (Figure 1B; all $p < 0.001$, one-sample t test). NHP T's choices were more binary, consistent with their overall higher performance (Figure 1C, top), whereas NHP I's choices showed a more graded dependence on the relative reward difference (Figure 1C, bottom). Nonetheless, all major effects were robust across both NHPs. Our analyses focus on trials in which NHPs chose the higher value target.

Not all trials required an explicit comparison between the reward value of the targets. If the reward value of the first target (R1) was the highest possible value, the NHPs could immediately decide to choose it. Alternatively, if R1 was the lowest possible value, the NHPs could proactively decide to choose the second target when it appeared. We instead focused neurophysiological analysis on the remainder of trials. On those, the NHPs had to wait until the second reward value cue (R2) before they could decide which target had the higher value. Thus, we focused on the time periods just before and after R2. After the second target (T2) but before its

analysis (LDA) classifier to decode each reward cue's value and each target's location from spike rates across the entire neural population. We found an increase of reward information (decoding accuracy) after each reward cue was presented and the reward information was maintained until NHPs made responses (Figure 2A). There was also information about other value-related variables, including chosen value and relative value (Figure S1). Next, we compared location information for chosen vs. unchosen targets separately for trials when the first vs. second target was chosen. Before the decision, both target locations could be decoded with similar accuracy. But after the second reward cue (R2), when the decision could be made, decoder accuracy for the location of the chosen target increased (Figures 2B and 2C). Information reflecting the first target location increased when it was chosen (Figure 2B), and there was a corresponding increase for the second target when it was chosen (Figure 2C). This resulted in an interaction between target order and choice ($p < 0.001$, bootstrap test on difference-of-differences; see STAR Methods). We also observed a small subject-specific modulation of this effect that paralleled differences in their behavior (Figure S2). NHP I was slightly biased toward choosing the first target and decoding accuracy difference between chosen and unchosen targets was also slightly greater

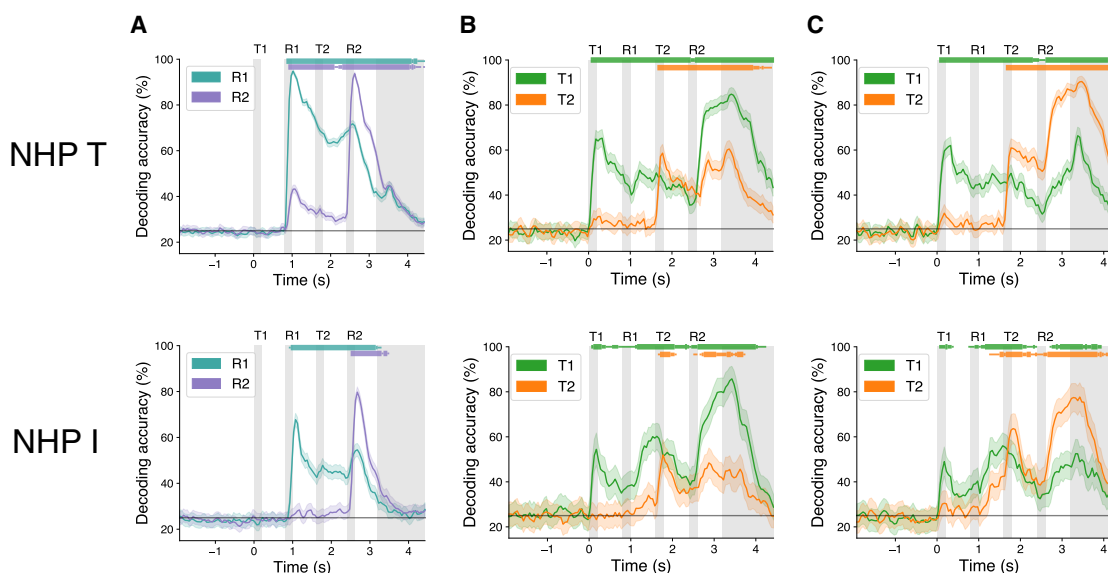


Figure 2. LPFC encoded information about both reward cue value and target location

(A) Decoding accuracy of first (R1; cyan) and second (R2; purple) reward cue value (mean $\pm$ SD of pseudopopulation bootstraps). (B and C) Decoding accuracy for the first (T1; green) and second (T2; orange) target location in trials where the first target was chosen (R1 > R2; B) and where the second target was chosen (R1 < R2; C) for both NHPs (mean $\pm$ SD of pseudopopulation bootstraps). Width of markers at top indicates significance: $p < 0.05$, $p < 0.01$, $p < 0.001$ for thin, medium, thick markers, respectively (corrected one-sided bootstrap against chance level).

when the first target was chosen, while choice behavior and neural information were both slightly biased toward the second target for NHP T.

The LDA classifier finds the axes in population activity space that maximize the ratio of between-class variance to within-class variance. Thus, the increases in accuracy of target information could reflect either increases in between-class variance or decreases in within-class variance. We found that the between-class variance increased after the decision for both chosen and unchosen targets. This decision-related increase in between-class variance was greater for the chosen targets than the unchosen targets (Figures 3A–3C; $p < 0.001$ in both R1 > R2 and R1 < R2 for both NHPs, chosen vs. unchosen targets bootstrap test), similar to the decoding accuracy. By contrast, the within-class variance did not decrease (Figures 3B–3D; for NHP T, $p = 0.403$ in R1 > R2 and $p = 0.398$ in R1 < R2; for NHP I, $p = 1$ in R1 > R2 and $p = 1$ in R1 < R2; pre- vs. post-decision bootstrap test). Thus, the increased decoding accuracy was due to increased separation of the representation of the target locations, not a decrease in their noise.

Decisions rotated chosen and unchosen targets into orthogonal subspaces

To examine how decisions transform neural subspaces, we first contrasted the representation of a given target (first/T1 or second/T2) between trials that resulted in different choices. For example, comparing the representation of the first target on trials when it was chosen (R1 > R2) vs. trials when the second target was chosen (R1 < R2; an analogous comparison was made for the second target; Figure 4A). To estimate the prefrontal population code for each target, we first fitted a Lasso regression for each neuron's spike rate using the loca-

tions of the two targets as regressors. We then used the fitted regression coefficients across neurons as the population coding vectors for the two target locations.²⁴ Using the fitted regression coefficients, rather than raw firing rates, to estimate population geometry eliminated any biases due to the task design (Figure S3). We visualized the representational geometry of compared target subspaces by projecting the population responses from an example pseudopopulation onto their top three principal components and quantified the alignment between the subspaces. To quantify the alignment between the subspaces in compared conditions, we computed Pearson's correlation between population coding vectors for corresponding target locations (Figure 4B).^{8,43,44} We then averaged correlations across the four target locations to obtain a final “subspace alignment” metric. This measure reflects the geometric alignment of both the compared subspaces and of the arrangement of conditions within each subspace, and it is similar to measures previously used for analogous purposes.^{8,38} This was performed separately at each time point before and after the decision.

We found that decisions resulted in an orthogonalization of the representation of chosen vs. unchosen targets. Before decisions could be made (before R2), the subspaces for the first target within trials where it would later be chosen vs. unchosen formed parallel planes where the representation of corresponding spatial locations was aligned (Figure 4C, left). After the decision, the subspaces of the chosen and unchosen first targets became near orthogonal (Figure 4C, right). Similar changes in representational geometry were observed when comparing chosen vs. unchosen second targets (Figure S4).

We quantified this change in population geometry by computing the subspace alignment between chosen and

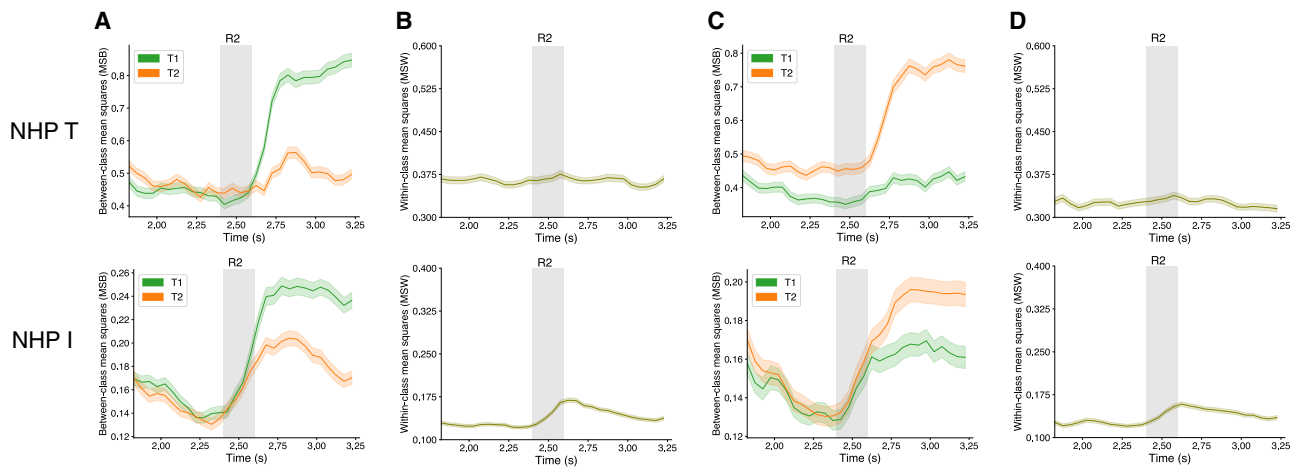


Figure 3. Between- and within-class variance of target information

(A and C) Between-class variance (mean $\pm$ SD of pseudopopulation bootstraps) of target locations for the first (green) and second (orange) targets, averaged over all neurons for trials where the first target was chosen (A) and where the second target was chosen (C).

(B and D) Within-class variance of target locations, averaged over all neurons for trials where the first target was chosen (B) and where the second target was chosen (D). The within-class variance has only a single curve because it was computed for each combination of first and second target locations (i.e., each “cell” in the $T1 \times T2$ design; see [STAR Methods](#)). After the decision, the representation of chosen target locations became more separated, resulting in increased decoding accuracy.

unchosen targets across time before and after the decision ([Figure 4D](#)). Alignment was positive before the decision, suggesting that pre-decision representations were similar for targets of the same order. Note that alignment values < 1 (perfect correlation) are expected, due to neural noise. Then the alignment significantly decreased ($p < 0.001$ for both NHPs, pre- vs. post-decision bootstrap test), to a value close to zero. This indicates that chosen and unchosen targets were kept in near-orthogonal subspaces, which minimizes the interference between them. The alignment between subspaces was also quantified using principal angles,⁴⁵ which showed qualitatively consistent results ([Figure S5A](#)). We also computed and compared the latency of value signals and changes in target geometry ([Figure 4D](#), arrows). We found that the value signal preceded the orthogonalization of chosen and unchosen targets, consistent with the geometry change being guided by value-based decisions.

The orthogonal representation of chosen and unchosen targets after the decision could in theory reflect two independent subpopulations of neurons. To test for this, we quantified the overlap between the neural populations with post-decision selectivity for the locations of chosen and unchosen targets ([Figure S6A](#)). For NHP T, 32% of neurons selective for the location of the chosen target and 74% of neurons selective for the location of the unchosen target showed selectivity for the locations of *both* targets. For NHP I, 30% of chosen-selective neurons and 74% of unchosen-selective neurons were selective for both targets. Examples of single neurons with distinct spatial tuning profiles for chosen vs. unchosen targets are shown in [Figure S6C](#). This indicates our results reflect different activity patterns for chosen and unchosen targets across a largely overlapping neural population, rather than separate subpopulations.

Decisions brought orthogonal subspaces into alignment for chosen targets

Next, we contrasted representations of conditions where targets had different temporal order but resulted in the same choice ([Figure 5A](#)). That is, we computed the alignment between the subspaces of the first and second targets on trials where each of them were *chosen*, and analogous comparisons between the first and second targets on trials where they were each *unchosen* ([Figure 5B](#)). Before the decision, the first (solid lines) and second (dashed lines) targets each formed approximately planar structures that were nearly orthogonal ([Figure 5C](#), left). After the decision, the chosen first and second targets (derived from separate trials) formed parallel, aligned planes ([Figure 5C](#), right). The alignment between the subspaces of the two targets before the decision was slightly positive but close to zero ([Figure 5D](#)). This indicates that the population activity patterns coding for the same set of target locations were initially nearly orthogonal for the first vs. second targets. After the decision, the alignment between the chosen first and second target subspaces (derived from separate trials; [Figure 5D](#), red lines) increased significantly ($p < 0.001$ for both NHPs, pre- vs. post-decision bootstrap test). In contrast, the alignment between the first and second targets when they were *unchosen* ([Figure 5D](#), blue lines) increased only slightly for one NHP (NHP T; $p < 0.001$) and showed no change for the other NHP (NHP I; $p = 0.866$). The alignment between subspaces was also quantified using principal angles, which showed qualitatively consistent results ([Figure S5B](#)). Again, the alignment between chosen targets followed the onset of the reward value signal, consistent with reward-based decision driving the alignment ([Figure 5D](#), arrows).

The observed alignment of chosen target representations could allow downstream areas to read out the location of the

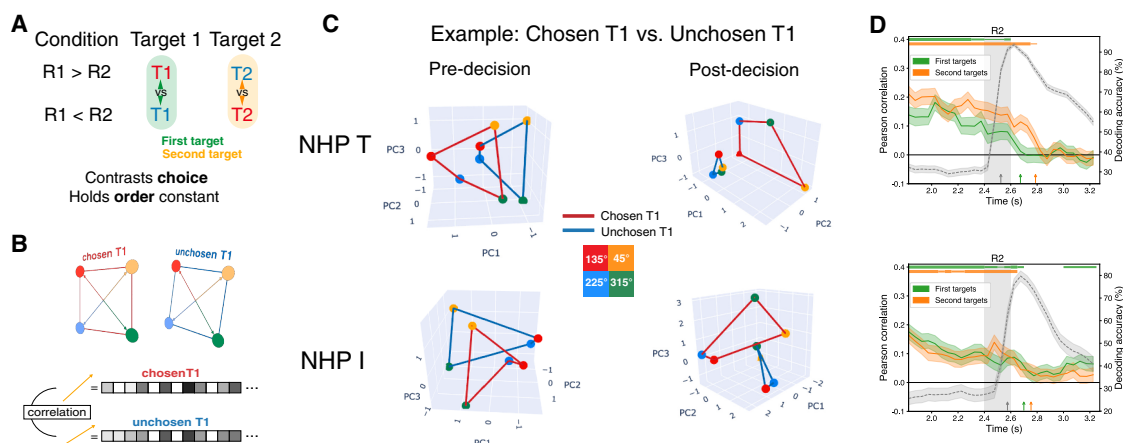


Figure 4. Chosen and unchosen targets became orthogonal after decision

(A) Schematic figure showing the contrast between representations of the same target (first/T1 or second/T2) conditions between trials that resulted in different choices (choose T1 [R1>R2] vs. choose T2 [R1<R2]).

(B) Subspace alignment between representation of chosen and unchosen first targets (or between chosen and unchosen second targets) was computed as the average Pearson correlation between population coding vectors of corresponding locations between the two targets.

(C) Geometry of chosen and unchosen (blue lines) first targets before (left) and after (right) decision in reduced 3D space. Four colored dots correspond to the four target locations for each condition. After decision, chosen and unchosen targets are rotated into orthogonal subspaces.

(D) Dynamics of subspace alignment (mean $\pm$ SD across pseudopopulation bootstraps) between chosen vs. unchosen first targets (green line) and between chosen vs. unchosen second targets (orange line) across time. The gray dashed line shows information reflecting the second reward value (R2; replotted from Figure 2A), for temporal comparison. Markers at top represent corrected one-sided bootstrap of Pearson correlation for first targets (green) and second targets (orange) against zero. Marker width indicates significance: $p < 0.05$, $p < 0.01$, $p < 0.001$ for thin, medium, thick, respectively. Arrows at bottom represent the latency of R2 information (gray arrow) and the decrease in correlation for the first target (green arrows) and for the second target (orange arrow).

chosen target with a single decoder, regardless of its initial presentation order. We tested this by training classifiers to decode the chosen target location on trials where the *first* target was chosen, then testing on trials where the *second* target was chosen, and vice versa (Figure S7). This cross-decoding was nearly as accurate as training and testing on the same target-choice condition, confirming the feasibility of readout using a single decoder. For the orthogonal subspaces before decisions, we confirmed that they are supported by an overlapping neural population selective for both first and second targets, rather than separate subpopulations (Figure S6B). For NHP T, 60% of neurons selective for T1 and 55% of neurons selective for T2 showed mixed selectivity for both targets. For NHP I, 59% of T1 neurons and 60% of T2 neurons showed mixed selectivity for both targets. Examples of single neurons with distinct spatial tuning profiles for first vs. second targets are shown in Figure S6D.

Before the decision, the LFPC population representation was organized in a format that kept the representations of the first and second targets independent. After the decision, it reorganized into a format in which the chosen location could be read out consistently, independent of whether it was presented first or second. Thus, the two contrasts (Figures 4 and 5) combined indicated a reorganization of target representation from order of presentation to choice.

Aligned target subspaces remained separable after choice

The alignment between chosen target subspaces could in theory reflect a transition to a purely spatial motor preparation code. Instead we found that post-decision population activity

also maintained an orthogonalized representation of which target was chosen. We trained an LDA classifier to decode choice information (whether the first or second target was chosen on each trial). We found that choice information increased after the second reward cue, and remained strong through the post-decision delay (Figure 6A). We confirmed this by projecting the population coding vectors of chosen targets onto the LDA-estimated choice axis. We found that the two chosen targets could be separated on the choice axis after R2 and remained separable throughout the post-decision delay (Figure 6B). The choice axis was orthogonal to both aligned target location subspaces (Figure 6C). These results are also consistent with the example post-decision population geometries shown in Figure 5C. These results suggest a post-decision transformation of population geometry that maintained target selection information in a format independent of the selected location, rather than a transition to a pure spatial motor-preparation signal.

Early subspace reorganization reflected early decisions

For the findings presented so far, we excluded the trials where the first reward cue (R1) signaled the highest or lowest possible value. In these trials, the NHPs could assume the second target's reward must be lower or higher in value, respectively, than the first target's reward. Thus, they could make an early decision without waiting for the second reward cue (R2). Here, we examined these previously excluded "early decision" trials. We hypothesized that if an NHP was able to reliably make an early decision, the observed decision-related changes of neural representation (increased information of chosen target and

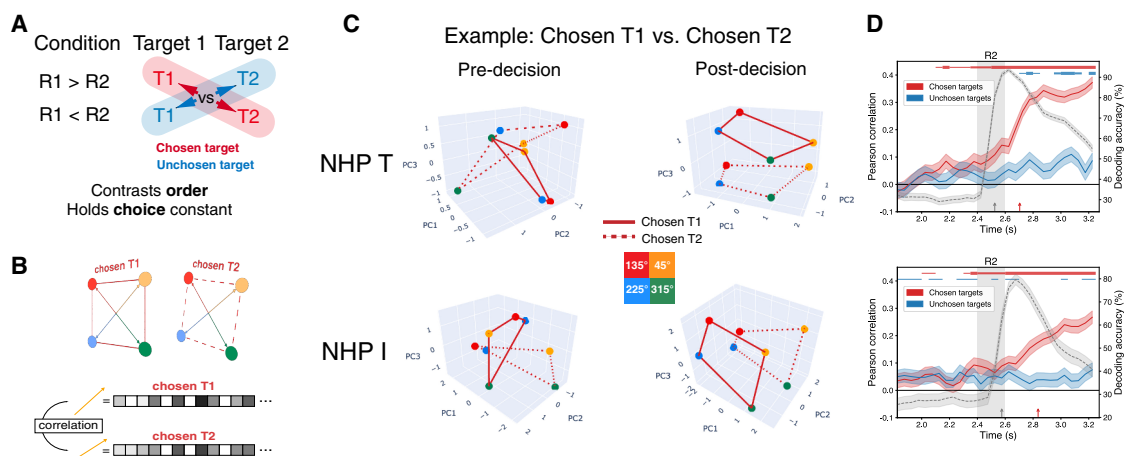


Figure 5. Target representations were reorganized by choice after decisions

(A) Schematic figure showing the contrast between targets with different order of presentation but resulting in the same choice. (B) Subspace alignment between first and second targets when they were each chosen (or between the two targets when they were both unchosen) was computed as the average Pearson correlation between population coding vectors of corresponding locations across the two targets. (C) Geometry of the first (solid lines) and second (dashed lines) targets when they were chosen, before (left) and after (right) decision in reduced 3D space. Four colored dots correspond to the four target locations. The first and second targets are initially held in orthogonal subspaces. After the decision, the first and second targets are rotated into the same, aligned subspace, but only when they are chosen (on separate trials). (D) Dynamics of subspace alignment (mean $\pm$ SD across pseudopopulation bootstraps) between first and second targets when they were each chosen (red line) and when they were each unchosen (blue line) across time for both NHPs. The gray dashed line represents the R2 information (replotted from Figure 2A). The markers at top represent corrected one-sided bootstrap of Pearson correlation for chosen targets (red) and unchosen targets (blue) against zero. Marker width indicates significance: $p < 0.05$, $p < 0.01$, $p < 0.001$ for thin, medium, thick, respectively. Arrows at bottom represent the latency of R2 information (gray arrow) and of the alignment between chosen targets (red arrows).

subspace reorganization) would also appear earlier in the trial than in the previously considered “late decision” trials.

We found mixed evidence for early decisions in the neural representations. For NHP T, information about the first target and second target increased when the first reward cue signaled the highest and lowest possible value, respectively (Figure S8A, solid green line; Figure S8B, solid orange line). NHP I showed a similar, but non-significant, trend only in trials where the first reward cue signaled the highest value (Figure S8C). The weaker and asymmetrical results in NHP I may be related to differences in their behavior between these conditions. When the first target had the highest possible value, NHP I performed well, regardless of the second value. When the first target had the lowest possible value, performance was poorer, especially when the second target was similar in value (Figure 1B). Thus, behavioral and neural results both suggest this subject may have made early decisions only when the first value was highest. These results offer some evidence that, when the NHPs can make early decisions, the increased information of chosen targets also occurs correspondingly early.

We also found evidence for early decisions in the decision-related neural subspace reorganization (orthogonalization and alignment; Figure S9). For NHP T, the early decision trials resulted in earlier orthogonalization of chosen vs. unchosen targets earlier relative to late-decision trials (Figures S9B and S9C, left), similar to the early changes in decoding accuracy. Similar, but weaker effects were observed in NHP I (Figures S9B and S9C, right). For both NHPs, there was also some evidence for early alignment of chosen target subspaces regardless of

the initial target presentation order on early-decision trials (Figure S9E). Also, the exact value of first reward (R1) had a graded effect on the alignment between first targets, such that targets with more similar values were more aligned (Figure S10). Taken together, these results provide evidence that, when early decisions are possible, the neural subspace reorganization reflecting decisions can happen correspondingly early.

DISCUSSION

We found that, during value-based decision-making, LPFC encoded both reward and option (i.e., target location) information and the representational geometry of options changed dynamically as the decision unfolded. When two potential options were simultaneously held in working memory, they were held in separate, near-orthogonal subspaces corresponding to their order of presentation. This presumably allowed the brain to keep the options separated by their behavioral context (i.e., order of presentation) before their relative values were known. After the choice could be made, neural subspaces of the prescribed actions rotated such that the representation of the chosen actions was aligned, regardless of their order of presentation. This shared subspace of chosen actions was near orthogonal to the subspaces of both unchosen actions, reducing interference from irrelevant information. In addition, options associated with higher values showed increased feature separation, which likely produces more accurate readout of the chosen action in downstream motor areas.

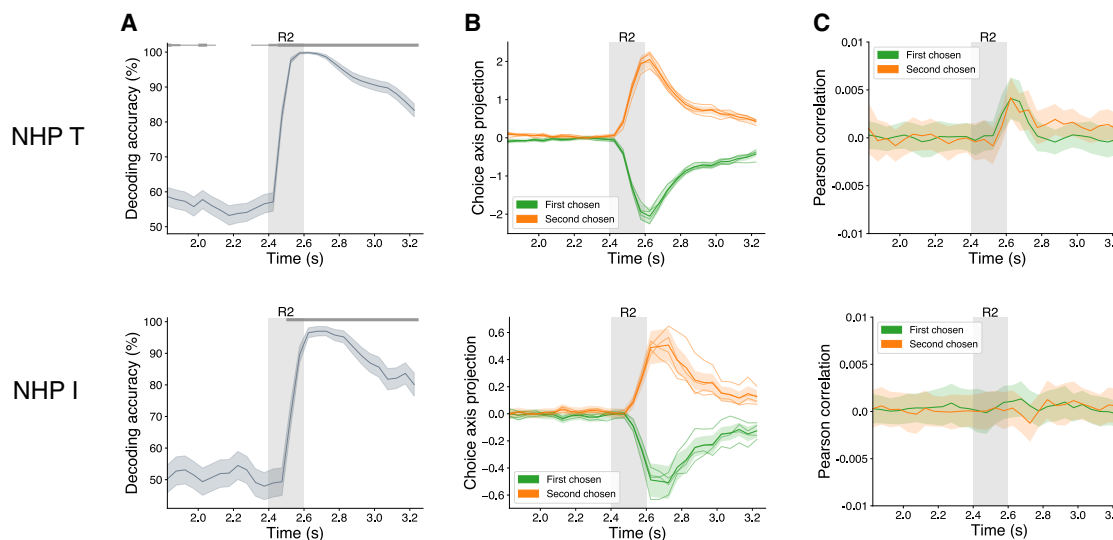


Figure 6. Aligned target subspaces remained separable by choice

(A) Decoding accuracy of choice (whether the first or second target was chosen) before and after decision for both NHPs (mean $\pm$ SD of pseudopopulation bootstraps). The markers at top represent corrected one-sided bootstrap of decoding accuracy of choice against chance level. Width of markers at top indicates significance: $p < 0.05$, $p < 0.01$, $p < 0.001$ for thin, medium, thick markers, respectively.

(B) The mean ($\pm$ SD pseudopopulation bootstraps) projection of population coding vectors of chosen targets (first target chosen: green line; second target chosen: orange line) onto the decoder choice axis. Lighter lines show the projection of each individual location for each chosen target.

(C) Subspace alignment (mean $\pm$ SD of pseudopopulation bootstraps) between choice axis and target location axes. Green line represents the alignment between choice axis and the first target chosen and orange line represents the alignment between choice axis and the second target chosen.

Orthogonalization of neural representations has been suggested as a mechanism to keep information independent.^{22,23,25,31,37–40,46–48} Neural activity changes reflecting one type of information would have no effect on other information held in an orthogonal subspace. Downstream processes could read out one with no interference from the other. Evidence has suggested mutual orthogonalization of sensory inputs and working memory,³⁷ of working memory sequences,²³ of perceptual choice and motor plans,⁴⁹ and of reach preparation and movement.³¹ Our results provide converging evidence for this idea in the context of two competing decision options. Orthogonalizing the two targets by order of presentation before the decision could keep their representations independent until it was time to select one. It may also support binding of the targets to their subsequent reward value assignments.⁵⁰ After the decision, keeping the representations of chosen and unchosen targets orthogonal may allow readout of the chosen target location without interference from the unchosen target.

Subspace alignment has been suggested to be important for discarding irrelevant information, while bringing relevant information into a common format to facilitate downstream readout.^{44,51–54} This could allow readout using a single linear decoder, regardless of any information irrelevant to the readout. In our study, the post-decision alignment of chosen targets onto a common subspace allows readout of target location using a single decoder regardless of their initial temporal order. This might, in part, reflect a transition to a neural code for motor preparation. However, importantly, information about the temporal order of the chosen target did not simply disappear. Instead, it was robustly maintained through the entire trial, but along a

population axis orthogonal to the aligned subspace coding for chosen target location. This suggests our results are better described by a transformation of population geometry that rotated the selected target into a common format suitable for downstream readout, rather than simply reflecting choice-independent preparatory motor activity. Preserving choice information in an orthogonal axis may enable LPFC to track the decision history and support later credit assignment without interfering with the readout of motor output.⁵⁵

The geometric changes we report largely reflect effects at the population level that are not easily reducible to effects at the single-neuron level. We found a reorganization from orthogonalization by target temporal order (first vs. second) to orthogonalization by choice status (chosen vs. unchosen). In both cases, the orthogonal subspaces largely reflected different patterns of activation across overlapping neural populations, rather than activation of separate subpopulations. This converges with analogous results across many different domains.^{31,38,56} It is also consistent with the well-established “mixed selectivity” of PFC neurons, where multiple task variables are encoded within overlapping neural populations and reorganized through changes in population activity patterns.^{26,44,57} In contrast, the increased population separation of chosen target locations could be explained by a gain increase at the single-neuron level. This is consistent with a large body of work showing enhanced single-neuron spiking activity and information content for behaviorally relevant information.^{58–64} However, we (and others^{35,65,66}) argue that, even in this case, population geometry provides a concise unifying analytical framework for understanding diverse neural effects,

which likely better approximates the level at which neural computation and communication actually happens in the brain.

Making decisions requires integrating various feature dimensions of each option into a representation of overall value, choosing the option with the highest value, then mapping the chosen option onto an appropriate action.^{1,2,67} Different brain regions have been implicated in different stages of this decision process. Evidence suggests orbital (OFC) and ventromedial PFC (vmPFC) encode an abstracted representation of overall option value and compare values of different options.^{9,68–73} Anterior cingulate cortex and motor-related areas such as supplementary eye field (SEF) engage in action selection and encode the chosen action in economic choices.^{74–76} A key region that can use the value signals to perform option selection and inform the action mapping is LPFC, which is involved in option selection and the mapping from chosen options onto appropriate actions.^{11,77,78}

Our results showed that LPFC may be involved in multiple stages of this decision process. First, LPFC holds multiple options online before decisions, within orthogonal subspaces. This resembles findings in orbital and medial PFC, where option values are encoded in orthogonal subspaces.⁵⁰ Second, LPFC induces a rotation in LPFC subspaces representing potential options and/or their associated actions, guided by the emerging value-based decisions. This is analogous to the choice-induced rotation of the OFC/vmPFC subspaces representing option values induced.⁵⁶ We further show the subspace rotation reflects a shift from organization by offer presentation order (first vs. second) to organization by choice status (chosen vs. unchosen). Previous work has shown that value signals can guide action selection by modulating the buildup of choice-related activity in ACC.⁷⁶ Our results provide a complementary perspective by showing that value-driven selection also reshapes the geometry of option subspaces, reorganizing subspaces from an order-based format into a choice-based format that supports action selection. Moreover, LPFC does not induce the decision-related changes in neural representation only after both values were revealed. Instead, it shapes target representations using the partial value information even before both values were revealed. Target information was enhanced when it was associated with highest value, and targets of more similar values occupied more aligned subspaces. This suggested that partial value information can begin to bias target representations before the full decision is available, thereby facilitating subsequent selection.

In summary, our results illustrate the dynamic subspace reorganization supporting option maintenance and selection in economic decisions. Before making the decisions, the brain maintained two possible options using orthogonalization. When it came time to make a decision, the decision induced a rotation of the chosen options into a common format, facilitating their translation into action. These findings highlight a role for dynamic changes in representational geometry in working memory and decision-making.

Limitations of the study

Several limitations should be considered when interpreting these findings. First, our task used a small, discrete set of spatial

locations and reward levels, and value was cued abstractly through color rather than being tied to natural, continuously varying stimulus features. This design was useful for isolating order- and choice-related geometry, but it leaves open whether the same reorganization would emerge with continuous value signals, larger choice sets, or more naturalistic decisions involving multiple attributes. Second, our analyses are correlational. Although the temporal relationship between value signals and subspace reorganization suggests that decisions drive these geometric transformations, further manipulations of LPFC activity may establish causality.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Earl K Miller (ekmiller@mit.edu).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- The neural data will be made available upon reasonable request to the [lead contact](#).
- Analysis scripts are provided on GitHub: <https://github.com/hdl1115/wmUpdate>
- Any additional information will be made available upon reasonable request to the [lead contact](#).

ACKNOWLEDGMENTS

We thank Will Dorrell, Mikael Lundqvist, and Jonatan Nordmark for helpful comments and discussions on this work. This work was supported by Office of Naval Research MURI N00014-23-1-2768 (E.K.M.), Army Research Office W911NF2410228 (E.K.M.), NEI 1R01EY033430-01A1 (E.K.M.), The Freedom Together Foundation (E.K.M.), The Picower Institute for Learning and Memory (E.K.M.), The Simons Center for the Social Brain (E.K.M.), NINDS Sustainable Transformation of Institutional Research Rigor (STIRR) Program (H.L.), and the Knut and Alice Wallenberg Foundation (N.C.).

AUTHOR CONTRIBUTIONS

J.R. and E.K.M. designed the experiments. J.R. collected the data. S.L.B. curated the data. H.L., N.C., and S.L.B. designed and carried out the data analysis. H.L., N.C., S.L.B., and E.K.M. wrote the manuscript, and E.K.M. supervised the study.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- [KEY RESOURCES TABLE](#)
- [EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS](#)
- [METHOD DETAILS](#)
 - Task paradigm
 - Electrophysiological recording
 - Behavioral analysis
 - Neural data processing
 - Linear regression
 - Subspace alignment

- Decoding analysis
- Single-neuron selectivity and population overlap analysis
- QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.isci.2026.117492>.

Received: February 11, 2026

Revised: June 1, 2026

Accepted: August 25, 2026

REFERENCES

1. Kable, J.W., and Glimcher, P.W. (2009). The Neurobiology of Decision: Consensus and Controversy. *Neuron* 63, 733–745. <https://doi.org/10.1016/j.neuron.2009.09.003>.
2. Padoa-Schioppa, C. (2011). Neurobiology of Economic Choice: A Good-Based Model. *Annu. Rev. Neurosci.* 34, 333–359. <https://doi.org/10.1146/annurev-neuro-061010-113648>.
3. Rangel, A., Camerer, C., and Montague, P.R. (2008). A framework for studying the neurobiology of value-based decision making. *Nat. Rev. Neurosci.* 9, 545–556. <https://doi.org/10.1038/nrn2357>.
4. Rangel, A., and Hare, T. (2010). Neural computations associated with goal-directed choice. *Curr. Opin. Neurobiol.* 20, 262–270. <https://doi.org/10.1016/j.conb.2010.03.001>.
5. Yoo, S.B.M., and Hayden, B.Y. (2018). Economic Choice as an Untangling of Options into Actions. *Neuron* 99, 434–447. <https://doi.org/10.1016/j.neuron.2018.06.038>.
6. Ballesta, S., Shi, W., Conen, K.E., and Padoa-Schioppa, C. (2020). Values encoded in orbitofrontal cortex are causally related to economic choices. *Nature* 588, 450–453. <https://doi.org/10.1038/s41586-020-2880-x>.
7. Ballesta, S., Shi, W., and Padoa-Schioppa, C. (2022). Orbitofrontal cortex contributes to the comparison of values underlying economic choices. *Nat. Commun.* 13, 4405. <https://doi.org/10.1038/s41467-022-32199-y>.
8. Kimmel, D.L., Elsayed, G.F., Cunningham, J.P., and Newsome, W.T. (2020). Value and choice as separable and stable representations in orbitofrontal cortex. *Nat. Commun.* 11, 3466. <https://doi.org/10.1038/s41467-020-17058-y>.
9. Strait, C.E., Blanchard, T.C., and Hayden, B.Y. (2014). Reward Value Comparison via Mutual Inhibition in Ventromedial Prefrontal Cortex. *Neuron* 82, 1357–1366. <https://doi.org/10.1016/j.neuron.2014.04.032>.
10. Seo, M., Lee, E., and Averbeck, B.B. (2012). Action Selection and Action Value in Frontal-Striatal Circuits. *Neuron* 74, 947–960. <https://doi.org/10.1016/j.neuron.2012.03.037>.
11. Cai, X., and Padoa-Schioppa, C. (2014). Contributions of Orbitofrontal and Lateral Prefrontal Cortices to Economic Choice and the Good-to-Action Transformation. *Neuron* 81, 1140–1151. <https://doi.org/10.1016/j.neuron.2014.01.008>.
12. Yim, M.Y., Cai, X., and Wang, X.J. (2019). Transforming the Choice Outcome to an Action Plan in Monkey Lateral Prefrontal Cortex: A Neural Circuit Model. *Neuron* 103, 520–532.e5. <https://doi.org/10.1016/j.neuron.2019.05.032>.
13. Kennerley, S.W., and Wallis, J.D. (2009). Reward-Dependent Modulation of Working Memory in Lateral Prefrontal Cortex. *J. Neurosci.* 29, 3259–3270. <https://doi.org/10.1523/JNEUROSCI.5353-08.2009>.
14. Petrides, M., and Pandya, D.N. (2006). Efferent association pathways originating in the caudal prefrontal cortex in the macaque monkey. *J. Comp. Neurol.* 498, 227–251. <https://doi.org/10.1002/cne.21048>.
15. Saleem, K.S., Miller, B., and Price, J.L. (2014). Subdivisions and connection networks of the lateral prefrontal cortex in the macaque monkey. *J. Comp. Neurol.* 522, 1641–1690. <https://doi.org/10.1002/cne.23498>.
16. Wallis, J.D. (2007). Orbitofrontal Cortex and Its Contribution to Decision-Making. *Annu. Rev. Neurosci.* 30, 31–56. <https://doi.org/10.1146/annurev.neuro.30.051606.094334>.
17. Lu, M.T., Preston, J.B., and Strick, P.L. (1994). Interconnections between the prefrontal cortex and the premotor areas in the frontal lobe. *J. Comp. Neurol.* 341, 375–392. <https://doi.org/10.1002/cne.903410308>.
18. Takada, M., Nambu, A., Hatanaka, N., Tachibana, Y., Miyachi, S., Taira, M., and Inase, M. (2004). Organization of prefrontal outflow toward frontal motor-related areas in macaque monkeys. *Eur. J. Neurosci.* 19, 3328–3342. <https://doi.org/10.1111/j.0953-816X.2004.03425.x>.
19. Takahara, D., Inoue, K.I., Hirata, Y., Miyachi, S., Nambu, A., Takada, M., and Hoshi, E. (2012). Multisynaptic projections from the ventrolateral prefrontal cortex to the dorsal premotor cortex in macaques – anatomical substrate for conditional visuomotor behavior. *Eur. J. Neurosci.* 36, 3365–3375. <https://doi.org/10.1111/j.1460-9568.2012.08251.x>.
20. Mante, V., Sussillo, D., Shenoy, K.V., and Newsome, W.T. (2013). Context-dependent computation by recurrent dynamics in prefrontal cortex. *Nature* 503, 78–84. <https://doi.org/10.1038/nature12742>.
21. Abbass, M., Corrigan, B., Johnston, R., Gulli, R., Sachs, A., Lau, J.C., and Martinez-Trujillo, J. (2025). Prefrontal cortex neuronal ensembles dynamically encode task features during associative memory and virtual navigation. *Cell Rep.* 44, 115124. <https://doi.org/10.1016/j.celrep.2024.115124>.
22. Weber, J., Iwama, G., Solbakk, A.K., Blenkmann, A.O., Larsson, P.G., Ivanovic, J., Knight, R.T., Endestad, T., and Helfrich, R. (2023). Subspace partitioning in the human prefrontal cortex resolves cognitive interference. *Proc. Natl. Acad. Sci. USA* 120, e2220523120. <https://doi.org/10.1073/pnas.2220523120>.
23. Xie, Y., Hu, P., Li, J., Chen, J., Song, W., Wang, X.J., Yang, T., Dehaene, S., Tang, S., Min, B., and Wang, L. (2022). Geometry of sequence working memory in macaque prefrontal cortex. *Science* 375, 632–639. <https://doi.org/10.1126/science.abm0204>.
24. Tian, Z., Chen, J., Zhang, C., Min, B., Xu, B., and Wang, L. (2024). Mental programming of spatial sequences in working memory in the macaque frontal cortex. *Science* 385, eadp6091. <https://doi.org/10.1126/science.adp6091>.
25. Tang, C., Herikstad, R., Parthasarathy, A., Libedinsky, C., and Yen, S.C. (2020). Minimally dependent activity subspaces for working memory and motor preparation in the lateral prefrontal cortex. *eLife* 9, e58154. <https://doi.org/10.7554/eLife.58154>.
26. Parthasarathy, A., Tang, C., Herikstad, R., Cheong, L.F., Yen, S.C., and Libedinsky, C. (2019). Time-invariant working memory representations in the presence of code-morphing in the lateral prefrontal cortex. *Nat. Commun.* 10, 4995. <https://doi.org/10.1038/s41467-019-12841-y>.
27. Rouzitalab, A., Boulay, C.B., Park, J., Martinez-Trujillo, J.C., and Sachs, A.J. (2023). Ensembles code for associative learning in the primate lateral prefrontal cortex. *Cell Rep.* 42, 112449. <https://doi.org/10.1016/j.celrep.2023.112449>.
28. Aoi, M.C., Mante, V., and Pillow, J.W. (2020). Prefrontal cortex exhibits multidimensional dynamic encoding during decision-making. *Nat. Neurosci.* 23, 1410–1420. <https://doi.org/10.1038/s41593-020-0696-5>.
29. Murray, J.D., Bernacchia, A., Roy, N.A., Constantinidis, C., Romo, R., and Wang, X.J. (2017). Stable population coding for working memory coexists with heterogeneous neural dynamics in prefrontal cortex. *Proc. Natl. Acad. Sci. USA* 114, 394–399. <https://doi.org/10.1073/pnas.1619449114>.
30. Churchland, M.M., Cunningham, J.P., Kaufman, M.T., Foster, J.D., Nuyujukian, P., Ryu, S.I., and Shenoy, K.V. (2012). Neural population dynamics during reaching. *Nature* 487, 51–56. <https://doi.org/10.1038/nature11129>.
31. Elsayed, G.F., Lara, A.H., Kaufman, M.T., Churchland, M.M., and Cunningham, J.P. (2016). Reorganization between preparatory and movement population responses in motor cortex. *Nat. Commun.* 7, 13239. <https://doi.org/10.1038/ncomms13239>.

32. Perich, M.G., Gallego, J.A., and Miller, L.E. (2018). A Neural Population Mechanism for Rapid Learning. *Neuron* 100, 964–976.e7. <https://doi.org/10.1016/j.neuron.2018.09.030>.
33. Sadtler, P.T., Quick, K.M., Golub, M.D., Chase, S.M., Ryu, S.I., Tyler-Kabara, E.C., Yu, B.M., and Batista, A.P. (2014). Neural constraints on learning. *Nature* 512, 423–426. <https://doi.org/10.1038/nature13665>.
34. Ruff, D.A., Ni, A.M., and Cohen, M.R. (2018). Cognition as a Window into Neuronal Population Space. *Annu. Rev. Neurosci.* 41, 77–97. <https://doi.org/10.1146/annurev-neuro-080317-061936>.
35. Ebitz, R.B., and Hayden, B.Y. (2021). The population doctrine in cognitive neuroscience. *Neuron* 109, 3055–3068. <https://doi.org/10.1016/j.neuron.2021.07.011>.
36. Gallego, J.A., Perich, M.G., Miller, L.E., and Solla, S.A. (2017). Neural Manifolds for the Control of Movement. *Neuron* 94, 978–984. <https://doi.org/10.1016/j.neuron.2017.05.025>.
37. Libby, A., and Buschman, T.J. (2021). Rotational dynamics reduce interference between sensory and memory representations. *Nat. Neurosci.* 24, 715–726. <https://doi.org/10.1038/s41593-021-00821-9>.
38. Panichello, M.F., and Buschman, T.J. (2021). Shared mechanisms underlie the control of working memory and attention. *Nature* 592, 601–605. <https://doi.org/10.1038/s41586-021-03390-w>.
39. Flesch, T., Juechems, K., Dumbalska, T., Saxe, A., and Summerfield, C. (2022). Orthogonal representations for robust context-dependent task performance in brains and neural networks. *Neuron* 110, 1258–1270.e11. <https://doi.org/10.1016/j.neuron.2022.01.005>.
40. Kaufman, M.T., Churchland, M.M., Ryu, S.I., and Shenoy, K.V. (2014). Cortical activity in the null space: permitting preparation without movement. *Nat. Neurosci.* 17, 440–448. <https://doi.org/10.1038/nn.3643>.
41. Semedo, J.D., Jasper, A.I., Zandvakili, A., Krishna, A., Aschner, A., Machens, C.K., Kohn, A., and Yu, B.M. (2022). Feedforward and feedback interactions between visual cortical areas use different population activity patterns. *Nat. Commun.* 13, 1099. <https://doi.org/10.1038/s41467-022-28552-w>.
42. MacDowell, C.J., Libby, A., Jahn, C.I., Tafazoli, S., Ardalan, A., and Buschman, T.J. (2025). Multiplexed subspaces route neural activity across brain-wide networks. *Nat. Commun.* 16, 3359. <https://doi.org/10.1038/s41467-025-58698-2>.
43. Stokes, M.G., Kusunoki, M., Sigala, N., Nili, H., Gaffan, D., and Duncan, J. (2013). Dynamic coding for cognitive control in prefrontal cortex. *Neuron* 78, 364–375. <https://doi.org/10.1016/j.neuron.2013.01.039>.
44. Bernardi, S., Benna, M.K., Rigotti, M., Munuera, J., Fusi, S., and Salzman, C.D. (2020). The Geometry of Abstraction in the Hippocampus and Prefrontal Cortex. *Cell* 183, 954–967.e21. <https://doi.org/10.1016/j.cell.2020.09.031>.
45. Gallego, J.A., Perich, M.G., Naufel, S.N., Ethier, C., Solla, S.A., and Miller, L.E. (2018). Cortical population activity within a preserved neural manifold underlies multiple motor behaviors. *Nat. Commun.* 9, 4233. <https://doi.org/10.1038/s41467-018-06560-z>.
46. Bouchacourt, F., and Buschman, T.J. (2019). A Flexible Model of Working Memory. *Neuron* 103, 147–160.e8. <https://doi.org/10.1016/j.neuron.2019.04.020>.
47. Nogueira, R., Rodgers, C.C., Bruno, R.M., and Fusi, S. (2023). The geometry of cortical representations of touch in rodents. *Nat. Neurosci.* 26, 239–250. <https://doi.org/10.1038/s41593-022-01237-9>.
48. Ritz, H., and Shenhav, A. (2024). Orthogonal neural encoding of targets and distractors supports multivariate cognitive control. *Nat. Hum. Behav.* 8, 945–961. <https://doi.org/10.1038/s41562-024-01826-7>.
49. Charlton, J.A., and Goris, R.L.T. (2024). Abstract deliberation by visuomotor neurons in prefrontal cortex. *Nat. Neurosci.* 27, 1167–1175. <https://doi.org/10.1038/s41593-024-01635-1>.
50. Johnston, W.J., Fine, J.M., Yoo, S.B.M., Ebitz, R.B., and Hayden, B.Y. (2024). Semi-orthogonal subspaces for value mediate a binding and generalization trade-off. *Nat. Neurosci.* 27, 2218–2230. <https://doi.org/10.1038/s41593-024-01758-5>.
51. Luyckx, F., Nili, H., Spitzer, B., and Summerfield, C. (2019). Neural structure mapping in human probabilistic reward learning. *eLife* 8, e42816. <https://doi.org/10.7554/eLife.42816>.
52. Morton, N.W., Schlichting, M.L., and Preston, A.R. (2020). Representations of common event structure in medial temporal lobe and frontoparietal cortex support efficient inference. *Proc. Natl. Acad. Sci. USA* 117, 29338–29345. <https://doi.org/10.1073/pnas.1912338117>.
53. Muhle-Karbe, P.S., Sheahan, H., Pezzulo, G., Spiers, H.J., Chien, S., Schuck, N.W., and Summerfield, C. (2023). Goal-seeking compresses neural codes for space in the human hippocampus and orbitofrontal cortex. *Neuron* 111, 3885–3899.e6. <https://doi.org/10.1016/j.neuron.2023.08.021>.
54. Sheahan, H., Luyckx, F., Nelli, S., Teupe, C., and Summerfield, C. (2021). Neural state space alignment for magnitude generalization in humans and recurrent networks. *Neuron* 109, 1214–1226.e8. <https://doi.org/10.1016/j.neuron.2021.02.004>.
55. Asaad, W.F., Lauro, P.M., Perge, J.A., and Eskandar, E.N. (2017). Prefrontal Neurons Encode a Solution to the Credit-Assignment Problem. *J. Neurosci.* 37, 6995–7007. <https://doi.org/10.1523/JNEUROSCI.3311-16.2017>.
56. Yoo, S.B.M., and Hayden, B.Y. (2020). The Transition from Evaluation to Selection Involves Neural Subspace Reorganization in Core Reward Regions. *Neuron* 105, 712–724.e4. <https://doi.org/10.1016/j.neuron.2019.11.013>.
57. Rigotti, M., Barak, O., Warden, M.R., Wang, X.J., Daw, N.D., Miller, E.K., and Fusi, S. (2013). The importance of mixed selectivity in complex cognitive tasks. *Nature* 497, 585–590. <https://doi.org/10.1038/nature12160>.
58. Buschman, T.J., and Kastner, S. (2015). From Behavior to Neural Dynamics: An Integrated Theory of Attention. *Neuron* 88, 127–144. <https://doi.org/10.1016/j.neuron.2015.09.017>.
59. Desimone, R., and Duncan, J. (1995). Neural Mechanisms of Selective Visual Attention. *Annu. Rev. Neurosci.* 18, 193–222. <https://doi.org/10.1146/annurev.ne.18.030195.001205>.
60. Hasegawa, R.P., Matsumoto, M., and Mikami, A. (2000). Search Target Selection in Monkey Prefrontal Cortex. *J. Neurophysiol.* 84, 1692–1696. <https://doi.org/10.1152/jn.2000.84.3.1692>.
61. McAdams, C.J., and Reid, R.C. (2005). Attention Modulates the Responses of Simple Cells in Monkey Primary Visual Cortex. *J. Neurosci.* 25, 11023–11033. <https://doi.org/10.1523/JNEUROSCI.2904-05.2005>.
62. Reynolds, J.H., Pasternak, T., and Desimone, R. (2000). Attention Increases Sensitivity of V4 Neurons. *Neuron* 26, 703–714. [https://doi.org/10.1016/S0896-6273\(00\)81206-4](https://doi.org/10.1016/S0896-6273(00)81206-4).
63. Reynolds, J.H., and Heeger, D.J. (2009). The Normalization Model of Attention. *Neuron* 61, 168–185. <https://doi.org/10.1016/j.neuron.2009.01.002>.
64. Suzuki, M., and Gottlieb, J. (2013). Distinct neural mechanisms of distractor suppression in the frontal and parietal lobe. *Nat. Neurosci.* 16, 98–104. <https://doi.org/10.1038/nn.3282>.
65. Cheng, Y.A., Sanayei, M., Chen, X., Jia, K., Li, S., Fang, F., Watanabe, T., Thiele, A., and Zhang, R.Y. (2025). A neural geometry approach comprehensively explains apparently conflicting models of visual perceptual learning. *Nat. Hum. Behav.* 9, 1023–1040. <https://doi.org/10.1038/s41562-025-02149-x>.
66. Saxena, S., and Cunningham, J.P. (2019). Towards the neural population doctrine. *Curr. Opin. Neurobiol.* 55, 103–111. <https://doi.org/10.1016/j.conb.2019.02.002>.
67. Lee, D., Seo, H., and Jung, M.W. (2012). Neural Basis of Reinforcement Learning and Decision Making. *Annu. Rev. Neurosci.* 35, 287–308. <https://doi.org/10.1146/annurev-neuro-062111-150512>.
68. Kennerley, S.W., Dahmubed, A.F., Lara, A.H., and Wallis, J.D. (2009). Neurons in the Frontal Lobe Encode the Value of Multiple Decision Variables.

- J. Cogn. Neurosci. 21, 1162–1178. <https://doi.org/10.1162/jocn.2009.21100>.
69. Lim, S.L., O'Doherty, J.P., and Rangel, A. (2011). The Decision Value Computations in the vmPFC and Striatum Use a Relative Value Code That is Guided by Visual Attention. *J. Neurosci.* 31, 13214–13223. <https://doi.org/10.1523/JNEUROSCI.1246-11.2011>.
70. Padoa-Schioppa, C., and Assad, J.A. (2006). Neurons in the orbitofrontal cortex encode economic value. *Nature* 441, 223–226. <https://doi.org/10.1038/nature04676>.
71. Roesch, M.R., and Olson, C.R. (2005). Neuronal Activity in Primate Orbitofrontal Cortex Reflects the Value of Time. *J. Neurophysiol.* 94, 2457–2471. <https://doi.org/10.1152/jn.00373.2005>.
72. Tremblay, L., and Schultz, W. (1999). Relative reward preference in primate orbitofrontal cortex. *Nature* 398, 704–708. <https://doi.org/10.1038/19525>.
73. Wunderlich, K., Rangel, A., and O'Doherty, J.P. (2010). Economic choices can be made using only stimulus values. *Proc. Natl. Acad. Sci. USA* 107, 15005–15010. <https://doi.org/10.1073/pnas.1002258107>.
74. Chen, X., and Stuphorn, V. (2015). Sequential selection of economic good and action in medial frontal cortex of macaques during value-based decisions. *eLife* 4, e09418. <https://doi.org/10.7554/eLife.09418>.
75. Sul, J.H., Jo, S., Lee, D., and Jung, M.W. (2011). Role of rodent secondary motor cortex in value-based action selection. *Nat. Neurosci.* 14, 1202–1208. <https://doi.org/10.1038/nn.2881>.
76. Balewski, Z.Z., Elston, T.W., Knudsen, E.B., and Wallis, J.D. (2023). Value dynamics affect choice preparation during decision-making. *Nat. Neurosci.* 26, 1575–1583. <https://doi.org/10.1038/s41593-023-01407-3>.
77. Kobayashi, S., Nomoto, K., Watanabe, M., Hikosaka, O., Schultz, W., and Sakagami, M. (2006). Influences of Rewarding and Aversive Outcomes on Activity in Macaque Lateral Prefrontal Cortex. *Neuron* 51, 861–870. <https://doi.org/10.1016/j.neuron.2006.08.031>.
78. Wallis, J.D., and Miller, E.K. (2003). Neuronal activity in primate dorsolateral and orbital prefrontal cortex during performance of a reward preference task. *Eur. J. Neurosci.* 18, 2069–2081. <https://doi.org/10.1046/j.1460-9568.2003.02922.x>.
79. Meyers, E.M., and Kreiman, G. (2012). Tutorial on pattern classification in cell recording. In *Visual Population Codes*, N. Kriegeskorte and G. Kreiman, eds. (MIT Press), pp. 517–538.
80. Efron, B., and Tibshirani, R.J. (1994). An Introduction to the Bootstrap (Chapman and Hall/CRC). <https://doi.org/10.1201/9780429246593>. <https://www.taylorfrancis.com/books/9781000064988>.
81. Dobs, K., Yuan, J., Martinez, J., and Kanwisher, N. (2023). Behavioral signatures of face perception emerge in deep neural networks optimized for face recognition. *Proc. Natl. Acad. Sci. USA* 120, e2220642120. <https://doi.org/10.1073/pnas.2220642120>.
82. Fukuda, H., Ma, N., Suzuki, S., Harasawa, N., Ueno, K., Gardner, J.L., Ichinoh, N., Haruno, M., Cheng, K., and Nakahara, H. (2019). Computing Social Value Conversion in the Human Brain. *J. Neurosci.* 39, 5153–5172. <https://doi.org/10.1523/JNEUROSCI.3117-18.2019>.
83. Benjamini, Y., and Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. R. Stat. Soc. B Stat. Methodol.* 57, 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental models: Organisms/strains		
<i>Macaca mulatta</i>	Covance	RRID: NCBITaxon_9544
Software and algorithms		
Python 3	Ananconda	3.11; RRID: SCR_008394; https://www.anaconda.com/
scikit-learn	Ananconda	1.8.0; RRID: SCR_002577; https://www.anaconda.com/
statsmodels	Ananconda	0.14.6; RRID: SCR_016074; https://www.anaconda.com/
MATLAB	The Mathworks Inc.	R2023B; RRID: SCR_001622; https://www.mathworks.com/
Custom Scripts	This paper	https://github.com/hdl1115/wmUpdate

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Two adult macaque monkeys (*Macaca mulatta*), monkey T (male, 10 years old, 13 kg) and monkey I (female, 15 years old, 9 kg) served as subjects in this study. All procedures followed the guidelines of the Massachusetts Institute of Technology Committee on Animal Care (protocol number 0710–070–13) and the National Institutes of Health.

METHOD DETAILS

Task paradigm

We trained the NHP subjects on a value-based selection task (Figure 1A). In each trial, a sequence of two spatial targets (white squares; 200 ms each) appeared at distinct locations (four possible at polar angles of 45°, 135°, 225°, and 315°, all at 5° eccentricity). All 12 possible distinct first/second target location combinations were presented with equal probability. Following each spatial target (after a 600 ms delay), an abstract reward cue (200 ms) signaled the magnitude of juice reward that would be obtained by remembering and later directing a response toward the immediately preceding spatial target. The reward cues were non-spatially-informative annuli, with four possible colors reflecting distinct reward values offered. The colors of the four reward cues differed between subjects, to disentangle low-level visual features from the associated reward offer. The associated absolute reward values were also slightly different between subjects, to maintain comparable behavior. For NHP T, blue = 1 juice drop, yellow = 5 drops, red = 8 drops, and magenta = 10 drops. For NHP I, yellow = 1 juice drop, green = 6 drops, blue = 9 drops, and cyan = 13 drops. Each trial always contained two distinct spatial targets and two distinct reward cues—there were never ambiguous trials where the first and second cue signaled the same reward or where the first and second spatial locations were identical. Otherwise, spatial locations and reward cue values were set randomly and independently on each trial—any location could be paired with any offered reward value. Finally, response targets were displayed at the locations of both spatial targets from that trial. Subjects were allowed to choose one by making a saccadic response to either location and received the reward signaled by its associated reward cue. Note that either choice of response target was rewarded. So “correct choice” in this task means that the subject chose the target with the larger reward.

Electrophysiological recording

Recordings were conducted with multiple single-contact tungsten electrodes (FHC). Raw analog data was amplified and recorded at 30 kHz using the Blackrock Cerebus system. Spikes were extracted by band-pass filtering (250–7500 Hz) and manually setting online amplitude thresholds. Spikes were sorted manually offline into isolated single units (Plexon OfflineSorter). All units were included in analyses and no other preprocessing was performed on the spiking data. Behavior was controlled using the PsychToolbox (monkey T) or MonkeyLogic (monkey I) systems, and eye position was monitored at 1 kHz (Eyelink 1000).

Recordings were unilateral and spanned both ventrolateral and dorsolateral PFC (Brodman areas 46, 45, and 8A; Figure S11). Frontal Eye Field (FEF) sites were localized using standard microstimulation mapping—sites near the arcuate sulcus where low-current (<50 μ A) stimulation reliably evoked saccades. Removing the few putative FEF and premotor cortex sites produced nearly identical results (Figure S12).

The data includes 23 sessions from NHP T and 17 sessions for NHP I. Each session, we recorded from an average of 24 ± 2 (mean $\pm$ SD) electrodes for NHP T and 33 ± 10 electrodes for NHP I. This resulted in a total of 316 neurons from NHP T and 422 neurons from NHP I.

Behavioral analysis

All analyses described here and below were performed with custom codes written in Python. We computed the behavioral accuracy for each of the 12 combinations of the first and second reward offer. Note that there are only 12 reward combinations because the same reward cue was not repeated twice within a single trial.

To estimate the influence of reward difference on accuracy, we computed the probability of choosing the first spatial target as a function of reward difference (reward 1 – reward 2; resulting in six levels of reward difference = $\{-3, -2, -1, +1, +2, +3\}$) and we fitted the accuracy with a logistic function. Behavioral results (Figures 1B and 1C) are presented as means $\pm$ standard deviations across all sessions for each subject. Their significance was quantified using sessions as observations ($n = 23$ sessions for NHP T, 17 sessions for NHP I).

Neural data processing

All neural data analyses used only correct trials (where the subject chose the higher value target) and only sessions with at least 15 correct trials for each condition (target location 1 $\times$ target location 2 $\times$ choice) were included. For most analyses, we also only included trials in which the first reward does not have the highest or lowest values. In trials where the first reward had the highest or lowest value, subjects could immediately infer which target to choose, without making any comparison to the second reward value. Thus, we included the trials in which the first reward has intermediate reward levels and focused the analysis on the delay period before and after the second reward was presented, when a decision could be made.

For most neural analyses, we generated pseudo-simultaneous population responses (“pseudopopulations”) by combining neurons across sessions as if they were sampled simultaneously, subsampling trials to match their count for each condition across neurons.⁷⁹ We randomly resampled pseudopopulation trials 1000 times to estimate means and standard deviations of summary statistics, and to evaluate their statistical significance. We calculated spiking rates using a 150 ms sliding window with 50 ms step size.

Linear regression

To deal with the non-independence between the two target locations in our task design (i.e., the two spatial targets cannot be the same within a single trial), we used linear multiple regression to quantify how each neuron’s activity was independently affected by the two spatial targets, and used the resulting fitted coefficients as our estimate of the neural population response.²⁴ We assumed that the average spiking rate $y_{i,t}$ of the i_{th} neuron at time t is the linear combination of effects due to the first and second spatial targets, such that:

$$y_{i,t} = \sum_{s=1}^2 \sum_{l=1}^4 \beta_{i,t}(s,l) X_{s,l} + \epsilon_{i,t}$$

Where $\beta(s,l)$ is the coefficient for location l (within the set $\{45^\circ, 135^\circ, 225^\circ, 315^\circ\}$) of target s (within the set $\{T1, T2\}$), $X_{s,l}$ is the regressor reflecting location l of target s (“dummy” encoded as $\{0, 1\}$), and ϵ_i is the residual error. For each trial, a sequence of two targets can be represented as an 8-dimensional vector and each target location is one-hot encoded. For example, sequence $[45^\circ, 135^\circ]$ can be represented as $[(1 \ 0 \ 0 \ 0), (0 \ 1 \ 0 \ 0)]^T$. We wanted to estimate coefficients for all possible location conditions, but this naturally resulted in an over-determined, rank-deficient design matrix. To solve rank deficiency and avoid overfitting, a Lasso regularization term was added to the linear regression model, which has previously been successfully used for analogous analytical situations.²⁴ The regularization amplitude α was selected using 5-fold cross-validation. We fitted Lasso models across a grid of candidate α values ranging from 1 to 10^{-5} and selected the α that minimized the cross-validated prediction error. This Lasso regression method successfully corrected for any bias due to the non-independent design, both for simulated data with known ground truth and for real data from example neurons (Figure S3), as expected from previous work.²⁴

At each time t , we concatenated the coefficients for location l of target s across neurons and we defined this vector of coefficients as the population response of location l of target s at time t . We separated trials based on whether the first reward was higher (first target chosen) or whether the second reward was higher (second target chosen) and performed separate linear regressions for these two conditions. Using the first-target-chosen trials for regression, the population response of the first target represents a chosen target, while the population response of the second target represents an unchosen target. Using the second-target-chosen trials for regression, the population response of the first target represents an unchosen target, while the population response of the second target represents a chosen target.

Subspace alignment

We computed the degree of alignment between the representations of two targets by computing the Pearson’s correlation between the mean population response vectors for the corresponding locations of the two targets (e.g., location 1 of the first target and location 1 of the second target) and averaging across the four locations. We refer to the resulting index as the “subspace alignment” between two representations. Alignment between subspaces was also quantified using principal angles.⁴⁵ We estimated a 2D subspace for each target representation by applying principal component analysis (PCA) to the condition-mean population activity and finding the first two principal components. We computed the principal angles between the two subspaces by performing singular value decomposition on the product of the two subspaces. The singular values correspond to the cosines of the principal angles

between the subspaces. We report the smallest principal angle as a summary measure of subspace alignment, with smaller angles indicating stronger alignment between the two target representations.

Two types of contrasts were made between targets. First, we compared conditions with the same target presentation order, but resulting in different choices (Figure 2). That is, we computed the subspace alignment between the first target when it was chosen vs. unchosen, and between the second target when it was chosen vs. unchosen. Second, we compared conditions with different target order, but resulting in the same choice. That is, we computed the subspace alignment between the first target when it was chosen vs. the second target when it was chosen, and between the first target when it was unchosen and the second target when it was unchosen.

Decoding analysis

We trained linear discriminant analysis (LDA) classifiers to decode the reward values, value-related variables (chosen value and relative value), and the location of the first and second spatial targets, separately within each time window. To equate the number of trials for each class, trials were randomly subsampled to match the class with the lowest trial count. For decoding the chosen reward value (Figure S1A), we excluded the lowest reward value, since in correct trials it was never chosen, and decoded the three remaining reward values. Within each of the 1000 randomly resampled pseudopopulation, we performed 5-fold cross-validation to compute the decoding accuracy. Before training the decoder at each fold, we z-scored the training and testing data for each neuron based on the mean and standard deviation across all trials of the training data. To reveal how decisions affect the target representations, we trained the classifier separately for trials in which the first reward was higher and for trials in which the second reward was higher. To examine whether the observed decision-related improvement in decoding accuracy was caused by increased between-class variance or decreased within-class variance, we computed between-class and within-class variance. Between-class variance for a spatial target is defined as the sum of squared differences between each location mean and the overall mean divided by (number of locations - 1). We quantified within-class variance as the variation around the mean neural response for each combination of the first and second target locations. In this case, within-class variance is defined as the sum of squared difference between individual response and the mean response in each location combination divided by (total number of neural responses - the total number of location combinations).

Single-neuron selectivity and population overlap analysis

To quantify single-neuron selectivity for each target, we pooled each neuron's spiking activity within the pre-decision or post-decision epoch and fitted a linear regression model in which the two target identities were included as categorical predictors, each with four possible levels. For each neuron and each target variable, the first level was omitted as the reference category, and an intercept term was included in the model. Thus, regression coefficients for the remaining levels quantified changes in firing rate relative to the reference level. A neuron was classified as selective for a given target if at least one regression coefficient associated with that target variable was statistically significant ($p < 0.05$; two-tailed t test).

QUANTIFICATION AND STATISTICAL ANALYSIS

Unless otherwise noted, bootstrap-like resampling of pseudopopulation trials was used for all error quantification and statistical testing.⁸⁰ Plots show the mean and standard deviation of a measure (e.g., subspace alignment, decoding accuracy) across 1000 random bootstrap-like resamples of trials to form a pseudopopulation, as detailed above. The standard deviation of a summary statistic across bootstraps is an estimate of the standard error of the statistic.⁸⁰ Thus, plotted error bars represent a bootstrap estimate of the standard error of the mean. For statistical testing, p values were generated by counting the proportion of bootstrap samples less than the value of a statistic expected under the null distribution (one-sided test that statistic is greater than expected by chance). For comparisons before vs. after decisions, the statistic was the paired difference between values before (1.8–2.4s) and after (2.6–3.2s) the decision point (R2). To evaluate the target order $\times$ choice interaction effect on decoding accuracy (Figures 4A and 4B), we computed a “difference-of-differences” statistic commonly used for bootstrap tests of interactions.^{81,82} For each bootstrap sample, we computed the difference in decoding accuracy between the two target orders ($T1 - T2$) and between the two choice conditions (choose $T1 -$ choose $T2$), and then computed the difference of these differences ($(T1 - T2) - (\text{choose } T1 - \text{choose } T2)$). The distribution of this difference-of-differences statistic across bootstrap resamples was evaluated against a null value of 0. For statistics evaluated at multiple individual timepoints, p values were corrected for multiple comparisons across timepoints using the Benjamini/Hochberg procedure for controlling the false-discovery rate.⁸³